

5.1.1 MSTR

- **Name:**Master Register0
- **Description:**Master Register0
- **Size:**32 bits
- **Offset:**0x0
- **Exists:**Always

31:30	device_config
29	frequency_mode
x:24	active_ranks
23	Rsvd
22	frequency_ratio
21:20	active_logical_ranks
19:16	burst_rdwrr
15	dll_off_mode
14	Rsvd
13:12	data_bus_width
11	geardown_mode
10	en_2t_timing_mode
9	burstchop
8	burst_mode
7:6	Rsvd
5	lpddr4
4	ddr4
3	lpddr3
2	lpddr2
1	mobile
0	ddr3

Table 5-5 Fields for Register: MSTR

Bits	Name	Memory Access	Description
31:30	device_config	R/W	Indicates the configuration of the device used in the system. ■ 00 - x4 device ■ 01 - x8 device ■ 10 - x16 device ■ 11 - x32 device Value After Reset:0x0 Exists:MEMC_DDR4==1 Programming Mode:Static
29	frequency_mode	R/W	Selects which registers are used. ■ 0 - Original registers ■ 1 - When UMCTL2_FREQUENCY_NUM=2: FREQ1 registers When UMCTL2_FREQUENCY_NUM>2: Chosen by MSTR2.target_frequency register. At boot time, this register must be set to 0 and the original register values are used. Value After Reset:0x0 Exists:UMCTL2_FAST_FREQUENCY_CHANGE==1 Programming Mode:Quasi-dynamic Group 2